SENG 31242 – System Design Project

SithCare – Mental Health Channelling App

System Requirements Specifications Draft

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Declaration

We, the undersigned members of Group 10, hereby declare that this Software Requirements Specification (SRS) document for SithCare is our own original work prepared in partial fulfilment of the requirements for SENG 31242 – System Design Project. All sources of information and references have been duly acknowledged. This document has not been submitted elsewhere for any academic or professional qualification.

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Abstract

SithCare is a web-based mental health channelling application designed specifically for the Sri Lankan context. It addresses the critical gap in accessible, private, and stigma-free mental health support by enabling patients to discover qualified mental health specialists, book online or in-person appointments, access telemedicine consultations, and receive guidance through an AI-powered triage chatbot (SithBot).

This Software Requirements Specification (SRS) captures all functional and non-functional requirements elicited from stakeholder interviews, competitive analysis of existing platforms (BetterHelp, Talkspace, Wysa, NalandaCare), and a review of Sri Lanka's National Mental Health Policy 2020–2030 and the WHO Mental Health Atlas 2024. The system supports distinct dashboards for patients, clinic

administrators, and platform administrators, with multi-clinic management, a live patient queue tracker, anonymous community support, volunteer listener sessions, and an emergency response subsystem.

Keywords: mental health, telemedicine, appointment booking, SRS, Sri Lanka, multi-clinic, SithCare, RBAC, SENG 31242.

Document Control and Revision History

Version	Date	Author(s)	Description	Status
0.1	2026-04-30	Group 10	Initial outline and scope definition	Draft
1.0	2026-05-03	Group 10	Full SRS draft – Sprint 2 Requirements Engineering. All chapters complete.	Draft – Pending Review

Table of Contents

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for SithCare, a mental health channelling web application intended for deployment in Sri Lanka. The document is prepared in accordance with the SENG 31242 – System Design Project guidelines issued by the Software Engineering Teaching Unit, University of Kelaniya, and follows the IEEE 830-1998 Recommended Practice for Software Requirements Specifications.

The primary audience of this document is: the project supervisor (Eng. Dr. Tiroshan Madushanka), the development team (Group 10), and any future stakeholders involved in the SENG 34213 development phase. This is a draft version and will be updated after questionnaire analysis, stakeholder interviews, and supervisor feedback are incorporated.

1.2 Scope

SithCare ("Care Beyond Words") is a web application that:

- Enables patients across Sri Lanka to discover, filter, and book appointments with verified mental health professionals.
- Provides clinic administrators with tools to manage doctor schedules, patient records, payment tracking, and communications.
- Provides a platform administrator with oversight of all registered clinics, user management, and compliance reporting.
- Includes a telemedicine module for video consultations, an AI triage chatbot (SithBot), an emergency support subsystem, and an anonymous community forum.
- Is localised for the Sri Lankan context: Sinhala, Tamil, and English language support; LKR currency; local emergency helplines; and PayHere payment gateway integration.

Out of Scope: native mobile applications (Android/iOS), integration with government electronic health record (EHR) systems, and insurance claim processing are explicitly excluded from this version.

1.3 Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
SDS	Software Design Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
RBAC	Role-Based Access Control
JWT	JSON Web Token – authentication mechanism
OTP	One-Time Password
MFA	Multi-Factor Authentication
LKR	Sri Lankan Rupee
API	Application Programming Interface
TLS	Transport Layer Security
AES	Advanced Encryption Standard
MVP	Minimum Viable Product
MoSCoW	Must-have, Should-have, Could-have, Won't-have (prioritisation framework)
PDPA	Personal Data Protection Act – Sri Lankan data protection legislation
SLMC	Sri Lanka Medical Council – statutory body for registering medical practitioners

SLPA	Sri Lanka Psychological Association – professional body for psychologists
CBT	Cognitive Behavioural Therapy
SithBot	The AI triage chatbot component of SithCare
Pseudonymous ID	A platform identifier that hides a user's real identity in community features
Telemedicine	Remote consultation using video/audio communication tools

1.4 References

- Ministry of Health, Sri Lanka. Mental Health Service Providers Directory 2023. https://www.health.gov.lk/
- Ministry of Health, Sri Lanka. National Mental Health Policy 2020–2030.
- Ministry of Health, Sri Lanka. Digital Health Blueprint 2023.
- World Health Organization. Mental Health Atlas 2024 Country Profile: Sri Lanka.
- NCBI Bookshelf (NBK610790). Sri Lanka mental healthcare access and help-seeking barriers.
- IEEE 830-1998 – Recommended Practice for Software Requirements Specifications.
- SENG 31242 System Design Project Course Guideline and Industrial Standards, University of Kelaniya.

1.5 Overview of Document

The remainder of this document is organised as follows:

- Chapter 2 – Overall Description: system context, user classes, operating environment, assumptions, and user personas.
- Chapter 3 – System Analysis: fact-finding summary, existing system analysis, competitive analysis, actors, use case list, use case descriptions, and activity diagram plan.
- Chapter 4 – Functional Requirements: complete numbered FR list with MoSCoW priority, source, and verification method.

- Chapter 5 – Non-Functional Requirements: performance, security, usability, reliability, scalability, and maintainability.
- Chapter 6 – Alternative Solutions and Feasibility Study.
- Appendices: glossary, fact-finding artefacts, risk register, MoSCoW scope table, and repository artefact map.

2. Overall Description

2.1 Product Perspective

SithCare operates as a standalone, centralised, role-based web application rather than a set of separate systems. A single platform manages shared services such as authentication, user roles, appointment scheduling, payments, communication, community moderation, and emergency support. Users access different dashboards according to their role. The system interfaces with the following external services:

External System	Purpose	Integration Method
PayHere	Sri Lankan payment gateway for online appointment payments in LKR	REST API / PayHere Checkout SDK
SMS Provider (Dialog/Mobitel)	OTP delivery, appointment reminders, queue notifications	SMS Gateway API
Email Service (SMTP/SendGrid)	Booking confirmations, password resets, notifications	SMTP / REST API
Video SDK (Jitsi / Daily.co)	Telemedicine video consultation rooms	JavaScript SDK / REST API
Google Maps / OpenStreetMap	Clinic location display and directions	Maps API
Ministry of Health Helpline Data	Emergency helpline numbers (static integration)	Static configuration

2.2 Product Functions

- Authentication and role-based access for patients, specialists,system admins.

- Patient onboarding, profile management, privacy consent, and Adult Mode for elderly users.
- Patient onboarding, profile management, privacy consent handling, and Adult Mode for elderly/low-literacy users.
- Specialist search/filtering by category, language, availability, clinic, and consultation type.
- Appointment booking, rescheduling, cancellation, confirmation, and status tracking with SMS/email notifications.
- Clinic and doctor availability management including multiple clinics/branches and emergency same-day slots.
- Telemedicine support through secure consultation links with one-tap phone fallback.
- Payments using local PayHere gateway, manual/cash payment tracking, and PDF receipt generation.
- Emergency support with helplines, urgent booking, and consent-based location sharing.
- Live patient queue tracker displaying current token number and sending proximity notifications.
- Anonymous community consultation using pseudonymous identities.

2.3 User Classes and Characteristics

User Class	Technical Proficiency	Primary Needs
Patient (Urban)	Moderate – smartphone/web literate	Private appointment booking, telemedicine, community support, resource library
Patient (Rural/Elderly)	Low – minimal digital literacy	Simplified Adult Mode, SMS reminders, in-person booking, queue tracking
Mental Health Specialist / Doctor	Moderate – clinical background	View patient list, write session notes, manage availability, start video sessions

Clinic Administrator	Moderate – administrative background	Schedule management, payment tracking, patient records, doctor coordination
Platform Administrator	High – technical background	Clinic onboarding, platform analytics, user management, compliance reporting
Volunteer Listener	Low to Moderate	Access volunteer portal, join listener sessions, view training materials
Community Moderator	Moderate	Moderate anonymous forum posts, flag harmful content, manage community
Emergency Helplines / Responders	External actor	Shown or contacted during emergency/crisis support flows

2.4 Operating Environment

Dimension	Details
Web Application	Deployed on cloud hosting (AWS / Google Cloud / DigitalOcean) with autoscaling capability
Browser Support	Chrome 110+, Firefox 110+, Safari 16+, Edge 110+
Network	Designed for Sri Lankan 4G LTE; telemedicine degrades gracefully on 3G (>1 Mbps minimum)
Database	PostgreSQL with schema-per-tenant isolation for clinic data
Server OS	Ubuntu 22.04 LTS
Localisation	Sinhala (Unicode), Tamil (Unicode), English
Future	Native Android/iOS mobile applications are out of scope for v1.0 but may be considered in a later phase

2.5 Design and Implementation Constraints

- The system must comply with Sri Lanka's data protection obligations under the proposed Personal Data Protection Act (PDPA).
- Patient mental health records must be encrypted at rest (AES-256) and in transit (TLS 1.3).
- The AI chatbot (SithBot) must not provide clinical diagnoses; all AI-generated suggestions must display a medical disclaimer.
- Payment processing must use PayHere or equivalent Sri Lankan certified payment gateway only.
- Emergency location sharing requires explicit user confirmation before activation on a per-session basis.
- Volunteers must not provide clinical or medical advice; the system must enforce this through UI constraints and a signed code of conduct.
- Telemedicine is optional; patients must always be able to use in-person appointment booking without a video component.
- The initial design must be feasible within the SENG 31242 design timeline and later development semester.
- All major requirements and design artefacts must be version-controlled through the project GitHub organisation.

2.6 Assumptions and Dependencies

ID	Type	Assumption / Dependency
A1	Assumption	Users have access to a smartphone, desktop, or internet-enabled device for online services.
A2	Assumption	Clinics and mental health specialists are willing to maintain availability and appointment information digitally.
A3	Assumption	Payment gateway, SMS/email, video SDK, and hosting services remain available and affordable for the project scope.
A4	Assumption	Stakeholder questionnaires and interviews will provide enough evidence to refine the requirements before SDS.

A5	Assumption	Mental health specialists registered on the platform hold valid professional licences verified during onboarding.
D1	Dependency	System communication depends on external SMS/email providers (Dialog, Mobitel, SendGrid).
D2	Dependency	Online payments depend on the PayHere payment gateway; PayHere sandbox is used for testing.
D3	Dependency	Telemedicine depends on video SDK/service reliability and regional latency (Southeast Asia servers required).
D4	Dependency	Emergency support depends on helpline contact information supplied by the Ministry of Health.
D5	Dependency	Emergency helpline numbers provided by MoH are assumed to remain static throughout the project lifecycle.

3. System Analysis

3.1 Fact-Finding Techniques Used

Technique	Session Details	Key Findings
Structured Interview	Conducted 5 separate interviews for mental patients to identify the pain points	Stigma is primary barrier; patients prefer anonymous search; clinics need payment tracking; elderly need simplified UX.
Online Questionnaire / Survey	Distributed to 40 respondents via Google Forms (2026-04-15 to 2026-04-22).	Analysis of the finding have been uploaded to the research repository accordingly
Survey for the mental health specialists	Conducted 6 interviews for mental health specialists and counsellors	Identified the pain points , actual requirements of the specialists for conducting the activities in a better way.
Document Analysis	WHO Mental Health Atlas 2024, MoH Directory 2023, National Mental Health Policy 2020–2030, Digital Health Blueprint 2023.	2.5 mental health workers per 100,000; significant treatment gap; government digital health push creates market opportunity.

3.2 Existing System Analysis

3.2.1 Current State and Pain Points

Mental health care access in Sri Lanka currently relies on outpatient department (OPD) queues at government hospitals, manual telephone bookings at private psychiatric clinics, and informal WhatsApp groups for community support. Key pain points identified from interviews and research:

- No centralised directory of licensed mental health professionals accessible to the public.
- Telephone-only booking creates stigma – fear of being overheard by family members or colleagues.
- No digital queue tracking; patients wait at clinic premises for 1–4 hours with no information about wait times.
- Payment records kept in paper ledgers; no digital receipts or payment history for patients.
- No telemedicine option for rural patients or those with mobility issues.
- Crisis support helplines are unknown to most patients; numbers are not widely publicised.

3.2.2 Competitive Analysis Summary

Feature	BetterHelp	Talkspace	Wysa	NalandaCare	SithCare
Multi-clinic management	✗	✗	✗	✗	✓
Sri Lanka localisation	✗	✗	Partial (AI)	Partial	✓ Full
Live queue tracker	✗	✗	✗	✗	✓
AI triage chatbot	✗	✗	✓	✗	✓
Emergency support	✗	✗	Basic	✗	✓ Full
Telemedicine	✓	✓	✗	✗	✓

Anonymous community	✗	✗	✓	✗	✓
PayHere / LKR payment	✗	✗	✗	✓	✓
Adult/Elderly mode	✗	✗	✗	✗	✓
Volunteer listener	✗	✗	✗	✗	✓

3.3 System Boundary and Actors

The system boundary of SithCare encompasses all interactions between the following actors and the platform:

Actor	Type	Role / Interaction
Patient	Primary	Registers, searches specialists, books appointments, joins consultation, uses emergency/community/support features.
Elderly Patient / Adult Mode User	Primary	Uses simplified interface for essential functions: booking, guidance, and emergency access.
Mental Health Specialist / Doctor	Primary	Manages profile, availability, appointments, telemedicine link, session notes, and consultation status.
Clinic Administrator	Primary	Manages clinic branches, schedules, queue, payments, and appointment approvals.
Platform Administrator	Primary	Manages users, roles, clinics, content, reports, fundraising approvals, and system settings.
Actor	**Type**	**Role / Interaction**
Volunteer Listener	Primary	Provides supportive listening sessions following code of conduct and confidentiality rules.
Community Moderator	Primary	Reviews reported community content, volunteer sessions, and recovery stories.

AI Chatbot (SithBot)	System Actor	Automated triage; interacts with patients to suggest specialist categories.
Emergency Helpline / Responder	External	Shown or contacted during emergency/crisis support flows (Sumithrayo 1926, CCCline).
Payment Gateway (PayHere)	External	Processes online payments and returns transaction status to SithCare.
SMS/Email Notification Service	External	Delivers appointment reminders, OTPs, and booking confirmations.
Video Consultation Service	External	Hosts and manages telemedicine video rooms.

3.4 Use Case List

ID	Use Case Name	Primary Actor(s)	Description
UC-01	Register and Login	Patient / Doctor / Admin / Volunteer	User creates account with OTP verification, logs in securely, and is routed to their role-based dashboard.
UC-02	Manage User Profile and Consent	Patient	Patient updates personal information, sets language preferences, manages privacy consent, and adds emergency contacts.
UC-03	Search Mental Health Specialist	Patient	Patient searches and filters mental health professionals by category, language, consultation mode, location, and availability.
UC-04	Book Appointment	Patient / Clinic Admin / Doctor	Patient selects a slot and books an appointment; system confirms the booking and notifies all relevant parties.
UC-05	Manage Doctor Availability	Doctor / Clinic Admin	Doctor or clinic admin defines, updates, or blocks available time slots across one or more clinic locations.

ID	Use Case Name	Primary Actor(s)	Description
UC-06	Approve / Reschedule Appointment	Clinic Admin / Doctor	Clinic staff reviews pending appointment requests and approves, rejects, or reschedules as required.
UC-07	Join Telemedicine Consultation	Patient / Doctor	Patient joins a secure video consultation via a doctor-generated room link; fallback phone call provided if video fails.
UC-08	Process Payment and Generate Receipt	Patient / Payment Gateway / Clinic Admin	Patient pays online (PayHere) or via cash; system updates payment status and generates a PDF receipt.
UC-09	Track Live Patient Queue	Patient / Clinic Admin	System displays the current token being served and the patient's position; clinic admin advances the queue.
UC-10	Request Emergency Support	Patient / Doctor / Emergency Helpline	Patient activates emergency button; system displays crisis options, initiates contact, and shares location if consented.
UC-11	Use Anonymous Community Consultation	Patient / Community Moderator	Patient posts general mental health concerns anonymously; community responds and moderator oversees safety.
UC-12	Complete AI Triage Questionnaire	Patient / AI Triage Module (SithBot)	Patient answers guided questions; SithBot recommends a specialist category with a mandatory non-diagnosis disclaimer.

3.5 Use Case Descriptions

UC-01 – Register and Login

Use Case ID	UC-01
Use Case Name	Register and Login
Actor(s)	Patient, Doctor, Clinic Admin, Volunteer, Platform Admin
Preconditions	User is not logged in. User has a valid email address or phone number.

Postconditions – Success	Authenticated session created; user routed to correct role-based dashboard.
Postconditions – Failure	Account not created or login rejected; appropriate error message displayed.
Main Flow	1. User opens SithCare and navigates to /register or /login.
	2. For registration: user selects role, enters full name, email, phone, date of birth, preferred language, and password. 3. System validates input (email format, password strength, phone format). 4. System checks for duplicate email/phone. 5. System sends OTP to registered email/phone. 6. User enters OTP, System verifies OTP, activates account, and redirects to profile completion wizard. 7. For login: user enters credentials, system verifies role and account status, creates a secure JWT session, and redirects to dashboard.
Alternative Flow 1	At step 2 (Registration): If password is weak, system highlights requirements (min 8 chars, 1 uppercase, 1 digit, 1 special character). User corrects and resubmits.
Alternative Flow 2	At step 5: If email is already registered, system displays a generic message to prevent user enumeration. OTP resend is limited to 3 attempts per hour.
Alternative Flow 3	At step 8 (Login): Invalid credentials or suspended account – system displays error. After 5 failed attempts, account is temporarily locked for 15 minutes.
Business Rules	BR-01: OTPs expire after 10 minutes. BR-02: Max 3 OTP resend attempts per hour. BR-03: Min password: 8 chars, 1 uppercase, 1 digit, 1 special character. BR-04: JWT access tokens expire after 15 minutes; refresh tokens expire after 7 days.

UC-02 – Manage User Profile and Consent

Use Case ID	UC-02
Use Case Name	Manage User Profile and Consent
Actor(s)	Patient

Preconditions	Patient is registered and logged in to their dashboard.
Postconditions – Success	Patient profile is updated with the latest personal information, consent preferences, and emergency contact details.
Postconditions – Failure	Profile changes not saved; validation error message displayed; existing data remains unchanged.
Main Flow	1. Patient navigates to 'My Profile' from the dashboard. 2. Patient views current profile information: name, contact details, date of birth, preferred language, and profile photo. 3. Patient edits desired fields and clicks 'Save Changes'. 4. System validates inputs (e.g., phone number format, valid date of birth). 5. System saves updated profile and displays a success confirmation. 6. Patient navigates to 'Privacy & Consent' settings.
	7. System displays privacy policy and current consent status. 8. Patient reviews and updates consent preferences (e.g., data sharing for anonymous analytics). 9. Patient adds or updates emergency contact name and phone number. 10. System saves preferences and logs the consent update with a timestamp.
Alternative Flow 1	At step 4: If validation fails (e.g., invalid phone format), system highlights the specific field error and prevents saving until corrected.
Alternative Flow 2	At step 8: If patient withdraws previously given consent, system updates settings and queues data minimisation actions where applicable.
Alternative Flow 3	Patient enables Adult Mode from profile settings: system switches to simplified navigation, larger font size (minimum 18px), and reduced feature set for the session.
Business Rules	BR-05: Consent withdrawal is recorded with timestamp and user ID for audit purposes. BR-06: Emergency contact information is used only during emergency support flow (UC-10) and is not shared publicly. BR-07: Patients can upload a profile photo; file size must not exceed 2 MB; accepted formats are JPG and PNG.

UC-03 – Search Mental Health Specialist

Use Case ID	UC-03

Use Case Name	Search Mental Health Specialist
Actor(s)	Patient
Preconditions	Patient is logged in. At least one verified specialist is listed on the platform.
Postconditions – Success	A filtered list of matching specialists is displayed with profile summaries and available slots.
Postconditions – Failure	No matching specialists found; system displays 'No results found' with suggestions to broaden the search criteria.
Main Flow	1. Patient navigates to 'Find a Specialist' from the dashboard or the AI chatbot recommendation. 2. Patient enters search criteria using available filters: specialist category (e.g., CBT Therapist, Psychiatrist, Counsellor), preferred language (Sinhala, Tamil, English), consultation type (online/in-person), location/district, and availability date. 3. System queries the database and returns a ranked list of matching specialists. 4. System displays each specialist with: name, photo, category, SLMC/SLPA registration number, consultation fee in LKR, languages spoken, average rating, and next available slot.
	5. Patient clicks a specialist card to view the full profile. 6. Patient reviews full profile and clicks 'Book Appointment' to proceed to UC-04.
Alternative Flow 1	At step 3: If no specialists match all filters, system relaxes lower-priority filters (e.g., consultation type) and displays results with a note: 'Showing nearest matches – some filters were broadened.'
Alternative Flow 2	At step 4: If the patient was referred via the AI chatbot (UC-14), the recommended specialist category is pre-filled in the search filter.
Alternative Flow 3	Patient selects 'View on Map' to see clinic locations pinned on an interactive map (Google Maps / OpenStreetMap integration).
Business Rules	BR-08: Only verified specialists with status 'Active' appear in search results. BR09: Search results are sorted by availability (soonest slot first) by default. BR-10: Specialist SLMC/SLPA registration number must be verified by the Platform Admin before the profile is publicly listed (UC-15).

UC-04 – Book Appointment

Use Case ID	UC-04
Use Case Name	Book Appointment
Actor(s)	Patient, Clinic Appointment System, Payment Gateway (PayHere), SMS/Email Service
Preconditions	Patient is registered and logged in. Patient has selected a specialist from the search results (UC-03).
Postconditions – Success	Appointment confirmed; booking reference generated; SMS and email confirmation sent; appointment appears on patient and doctor dashboards.
Postconditions – Failure	Appointment not booked; selected slot remains available; patient notified with reason.
Main Flow	1. Patient selects a specialist and clicks 'Book Appointment'. 2. System displays available time slots for the selected clinic/doctor. 3. Patient selects a date and time slot and chooses consultation type (in-person or video). 4. Patient reviews booking summary: doctor name, clinic, slot, consultation type, and fee in LKR. 5. Patient selects payment method: online (PayHere) or cash at clinic. 6. If online: patient completes PayHere checkout; system receives payment confirmation. 7. System creates appointment record with status 'Confirmed'.
	8. System generates a unique booking reference number. 9. System sends SMS and email confirmation to patient. 10. Appointment appears on both patient and doctor dashboards.
Alternative Flow 1	At step 6: If PayHere payment fails, system retains the slot for 5 minutes and prompts retry. After 2 failures, slot is released and patient is redirected to search.
Alternative Flow 2	At step 2: If no regular slots are available, system offers 'Emergency Same-Day Booking' option and displays the earliest next available slot, subject to clinic approval.

Alternative Flow 3	Patient cancels before step 7: system releases the slot and no record is created.
Business Rules	BR-11: Appointment slots are 50 minutes by default. BR-12: Cancellation is free if more than 24 hours before the appointment; a cancellation fee policy applies within 24 hours. BR-13: Online payment receipt is generated as a PDF automatically upon payment confirmation. BR-14: Slot is reserved for 5 minutes during payment processing.

UC-05 – Manage Doctor Availability

Use Case ID	UC-05
Use Case Name	Manage Doctor Availability
Actor(s)	Doctor, Clinic Administrator
Preconditions	Doctor or clinic admin is authenticated and has the appropriate role permissions.
Postconditions – Success	Availability calendar is updated and reflects correctly in patient-facing specialist search.
Postconditions – Failure	Schedule conflict detected; changes not saved; user informed of the conflict.
Main Flow	1. Doctor or admin opens the availability management panel and selects a clinic/branch. 2. System displays a weekly calendar view of current availability. 3. User selects a date and defines available time slots (start time, end time, slot duration). 4. User can block specific dates/times for holidays or personal leave. 5. System validates for conflicts with existing confirmed appointments. 6. System saves the updated availability schedule. 7. Updated slots immediately become available in patient-facing search.
Alternative Flow 1	At step 5: If a conflict exists with an already-confirmed appointment, system highlights the conflict and prevents saving until resolved.
Alternative Flow 2	Doctor is linked to multiple clinics: system allows setting separate availability schedules per clinic branch.

Alternative Flow 3	Clinic admin sets availability on behalf of a doctor: system logs the change with the admin's account for audit purposes.
Business Rules	BR-15: A doctor may be associated with up to 5 clinic branches simultaneously. BR-16: Availability changes affecting already-confirmed appointments require explicit admin confirmation and trigger a patient notification.

UC-06 – Approve / Reschedule Appointment

Use Case ID	UC-06
Use Case Name	Approve / Reschedule Appointment
Actor(s)	Clinic Administrator, Doctor, SMS/Email Notification Service
Preconditions	One or more appointment requests with status 'Pending Approval' exist in the clinic admin dashboard.
Postconditions – Success	Appointment status updated (Confirmed/Rejected/Rescheduled); patient notified via SMS and email.
Postconditions – Failure	Action not taken; appointment remains in pending state; error displayed to admin.
Main Flow	1. Clinic admin logs in and navigates to 'Pending Appointments'. 2. System displays a list of pending appointment requests sorted by date. 3. Admin selects a request and reviews patient details, requested slot, and consultation type. 4. Admin chooses one of: Approve, Reject, or Reschedule. 5a. (Approve): System updates appointment status to Confirmed and sends confirmation to patient. 5b. (Reject): Admin enters a rejection reason. System updates status to Rejected and notifies patient with the reason and suggestion to re-book.
Alternative Flow 1	At step 5c: If patient does not confirm the rescheduled slot within 24 hours, the appointment is automatically cancelled and the patient is notified.

Alternative Flow 2	Emergency Same-Day Booking (from UC-04): admin is presented with a priority flag and must approve or reject within 30 minutes; a reminder notification is sent if not acted on within 15 minutes.
Business Rules	BR-17: Manual approval is required only for clinics that have enabled the 'Approval Required' setting. BR-18: Rejection reason is recorded for audit and reported to the Platform Administrator monthly.

UC-07 – Join Telemedicine Consultation

Use Case ID	UC-07
Use Case Name	Join Telemedicine Consultation
Actor(s)	Patient, Doctor, Video Consultation Service (Jitsi/Daily.co)
Preconditions	Appointment is confirmed with consultation type 'Online/Video'. Appointment time is within 15 minutes.
Postconditions – Success	Consultation session is started; both parties connected in the video room.
Postconditions – Failure	Video session fails; fallback phone call option is displayed to both parties.
Main Flow	1. System generates a unique, time-limited video room link 15 minutes before the appointment. 2. Patient opens the appointment in 'My Appointments' dashboard and clicks 'Join Session'. 3. System verifies appointment status and patient identity. 4. Patient is connected to the video room. 5. Doctor opens the appointment from the doctor dashboard and clicks 'Start Session'. 6. Doctor is connected; session status changes to 'In Progress'. 7. Upon completion, doctor clicks 'End Session'; appointment status updates to 'Completed'.
Alternative Flow 1	At step 2: Video link is expired or network failure occurs. System displays 'Unable to connect' with a one-tap phone call fallback button to call the doctor's registered phone number.

Alternative Flow 2	At step 4: Patient's device lacks camera/microphone permissions. System displays instructions for enabling permissions; audio-only fallback is offered.
Business Rules	BR-19: Video room links expire 30 minutes after the scheduled appointment end time. BR-20: Sessions are not recorded by the SithCare platform; recording is the doctor's responsibility if clinically required. BR-21: One-tap phone fallback must always be available if video fails.

UC-08 – Process Payment and Generate Receipt

Use Case ID	UC-08
Use Case Name	Process Payment and Generate Receipt
Actor(s)	Patient, Payment Gateway (PayHere), Clinic Administrator
Preconditions	Appointment requires payment. Patient has selected a payment method (online or cash).
Postconditions – Success	Payment status updated to Paid; PDF receipt generated and available to patient and clinic admin.
Postconditions – Failure	Payment not completed; appointment status remains Pending Payment; patient notified.

Main Flow	1. Patient views payment due on the appointment summary screen. 2. Patient selects payment method: Online (PayHere) or Cash at Clinic. 3a. (Online): System redirects to PayHere checkout page with appointment details and fee in LKR. PayHere processes payment and returns status to SithCare via webhook. 3b. (Cash): Clinic admin marks the appointment as Cash Paid after receiving payment at the clinic. 4. System updates appointment/payment record to 'Paid'. 5. System generates a PDF receipt with appointment details, fee, payment method, date, and booking reference. 6. Receipt is available for download from patient dashboard and clinic admin panel.
Alternative Flow 1	At step 3a: PayHere payment fails. System retains appointment slot for 5 minutes and prompts patient to retry. After 2 failures, slot is released.
Alternative Flow 2	Refund required (appointment cancelled within refund window): System initiates refund via PayHere API; refund status tracked and patient notified of completion.
Business Rules	BR-22: Payment card data is never stored on SithCare servers; all card handling is delegated to PayHere's PCI-DSS compliant infrastructure. BR-23: All amounts are displayed and processed in Sri Lankan Rupees (LKR). BR-24: PDF receipts are retained for 12 months and accessible to both patient and clinic admin.

UC-09 – Track Live Patient Queue

Use Case ID	UC-09
Use Case Name	Track Live Patient Queue
Actor(s)	Patient, Clinic Administrator
Preconditions	Patient has a confirmed in-person appointment for today with an assigned queue token. Clinic has queue tracking enabled.
Postconditions – Success	Patient can view their real-time queue position without being physically present at the clinic.

Postconditions – Failure	Queue data unavailable; static message displayed: 'Queue data temporarily unavailable. Please contact the clinic.'
Main Flow	1. Clinic admin starts the daily queue from the clinic dashboard. 2. System assigns each confirmed in-person patient a queue token number. 3. Patient logs in, navigates to 'My Appointments', and selects today's appointment. 4. System displays: current token being served, patient's token number, estimated wait time. 5. Queue tracker refreshes automatically every 60 seconds. 6. Clinic admin clicks 'Next Patient' to advance the queue. 7. System updates the display on all connected patient views in real time. 8. When patient is 3 tokens away, system sends a push/SMS notification: 'You are next soon – please head to the clinic.'
Alternative Flow 1	At step 5: If the clinic administrator pauses the queue (e.g., lunch break), a banner is shown: 'Queue is currently paused. Expected to resume at [time].'
Alternative Flow 2	Emergency slot is inserted by admin: queue is renumbered; affected patients receive an SMS notification of the delay.
Business Rules	BR-25: Queue position is updated only by the clinic administrator. BR-26: Patients are notified automatically when within 3 tokens of their turn. BR-27: Queue resets automatically at the end of each clinic day.

UC-10 – Request Emergency Support

Use Case ID	UC-10
Use Case Name	Request Emergency Support
Actor(s)	Patient, Doctor, Emergency Helpline
Preconditions	Patient is logged in and in distress. Emergency support button is visible on all screens.
Postconditions – Success	Patient is connected to assigned doctor or helpline; location shared (if consented); crisis event logged with timestamp.

Postconditions – Failure	Connection fails; patient is shown static helpline numbers with tap-to-call functionality.
Main Flow	1. Patient taps the emergency button (persistent red floating action button visible on all screens). 2. System displays: 'You are about to contact your assigned mental health professional. Your location will be shared. Confirm?' along with the disclaimer: SithCare does not replace emergency services. Dial 119 (police) or 1990 (ambulance) for immediate danger. 3. Patient taps Confirm. 4. System requests location permission from device (if not already granted). 5. System attempts to call the assigned doctor. 6. System logs the crisis event with timestamp and anonymised patient ID. 7. If doctor answers: patient is connected; consented location is shared with doctor.
Alternative Flow 1	At step 3: Patient taps Cancel. System returns to previous screen. No call placed; no location shared. No crisis event logged.
Alternative Flow 2	At step 4: Location services are disabled or permission denied. System skips location sharing, proceeds with call only, and informs the doctor that location was unavailable.
Business Rules	BR-28: Location sharing requires explicit per-session user consent. BR-29: Crisis events are logged for safeguarding review by the Platform Administrator. BR-30: SithCare must always display: 'SithCare is not a substitute for emergency services. Dial 119 for police or 1990 for ambulance.'

UC-11 – Use Anonymous Community Consultation

Use Case ID	UC-11
Use Case Name	Use Anonymous Community Consultation
Actor(s)	Patient, Community Moderator
Preconditions	Patient is logged in and has accepted community guidelines.

Postconditions – Success	General question is posted pseudonymously; community responses are visible; moderator can review if reported.
Postconditions – Failure	Post flagged and hidden; user informed of guideline violation.
Main Flow	1. Patient navigates to 'Community Forum' from the dashboard.
	2. System displays community safety guidelines. Patient accepts to proceed. 3. Patient submits a general mental health question using their pseudonymous platform identity. 4. System hides the real user identity & displays only the pseudonymous username. 5. Other community members (patients) can respond under their own pseudonymous identities. 6. Moderator monitors the forum and can flag, hide, or remove posts violating community guidelines.
Alternative Flow 1	Post contains private identifying information, clinical details, or emergency-level distress keywords: system warns user and recommends contacting a professional or using the emergency support feature.
Alternative Flow 2	User reports a post: moderator receives a review notification and must action within 24 hours.
Business Rules	BR-31: Real user identity is never publicly linked to community posts. BR-32: Community posts are not clinical advice; a disclaimer is permanently visible in the forum. BR-33: Moderators can escalate concerning posts to Platform Admin for further review.

UC-12 – Complete AI Triage Questionnaire (SithBot)

Use Case ID	UC-12
Use Case Name	Complete AI Triage Questionnaire (SithBot)
Actor(s)	Patient (or Unregistered Visitor), AI Triage Module (SithBot)
Preconditions	User has accessed the SithCare platform (login is not required for basic triage access).

Postconditions – Success	User receives a specialist category recommendation and is guided to book an appointment, access resources, or contact emergency support.
Postconditions – Failure	Chatbot cannot determine a recommendation; user is guided to contact the platform support team or browse resources manually.
Main Flow	1. User clicks 'Talk to SithBot' on the home or dashboard screen. 2. SithBot greets user and asks: 'How are you feeling today?' with quick-reply options (Anxious, Depressed, Stressed, Struggling, Not Sure). 3. User describes feelings or selects from options. 4. SithBot asks 2–4 follow-up questions based on the initial response. 5. SithBot analyses responses and suggests a specialist 6. SithBot presents options: Book Appointment | Browse Resources | Talk to a Volunteer | Emergency Support.
	7. Medical disclaimer is displayed: 'SithBot does not provide medical diagnoses. Please consult a qualified mental health professional.'
Alternative Flow 1	At step 3: User types crisis keywords (e.g., 'suicidal', 'harm myself', 'end my life'). SithBot immediately surfaces emergency support options and helpline numbers, skipping the triage flow entirely.
Alternative Flow 2	At step 5: Chatbot confidence is below threshold. SithBot responds: 'I want to make sure you get the right support. Let me connect you with a volunteer listener or a platform support contact.'
Business Rules	BR-40: SithBot must never use the word 'diagnose' or imply clinical assessment. BR-41: All chatbot interactions are logged in anonymised form for safety review by Platform Admin. BR-42: Medical disclaimer must appear on every SithBot response screen, not just the final screen.

3.6 Activity Diagram Plan

The following activity diagrams shall be created using Draw.io (UML 2.x notation) and committed to the repository under documents/diagrams/activity/ with both source (.drawio) and exported SVG/PDF files.

Diagram ID	Process / Use Case	Scope Description

AD-01	Patient Registration and OTP Verification (UC-01)	Account creation, input validation, OTP delivery, verification, and dashboard routing.
AD-02	Therapist Search and Appointment Booking (UC-03, UC-04)	Search filters, slot selection, payment flow, confirmation, and SMS/email notification.
AD-03	Clinic Appointment Approval and Queue Preparation (UC-06, UC-09)	Clinic admin approval, schedule update, queue token assignment, patient proximity notification.
AD-04	Emergency Support Request (UC-10)	Emergency button, disclaimer, consent confirmation, doctor call attempt, helpline fallback, location sharing, crisis log.
AD-05	Anonymous Community Consultation (UC-11)	Anonymous post creation, pseudonymous identity masking, moderation rules, community responses, and report handling.
AD-06	Payment Processing and Receipt Generation (UC-08)	Payment method selection, PayHere gateway redirect, webhook confirmation, record update, and PDF receipt generation.

3.7 High-Level System Sequence Descriptions

SSD ID	Name	Sequence Summary
SSD-01	Appointment Booking	Patient → SithCare: search specialist | SithCare → DB: retrieve slots | Patient → SithCare: confirm booking | SithCare → DB: create appointment | SithCare → PayHere: process payment | SithCare → Notification Service: send confirmation.
SSD-02	Telemedicine Consultation	Doctor → SithCare: generate room link | SithCare → Video SDK: create room | Patient → SithCare: join consultation | SithCare → Video SDK: validate and connect both parties.

SSD-03	Payment Processing	Patient → SithCare: initiate payment | SithCare → PayHere: create transaction | PayHere → SithCare: return status via webhook | SithCare → DB: update payment | SithCare → Patient: generate and deliver PDF receipt.
SSD-04	Emergency Support	Patient → SithCare: request emergency | SithCare → Patient: display disclaimer/options | Patient → SithCare: confirm + consent to location | SithCare → Doctor: initiate call | [fallback] SithCare → Patient: display helpline numbers.
SSD-05	AI Chatbot Triage	Patient → SithBot: describe feelings | SithBot → Patient: ask followup questions | SithBot → Recommendation Engine: process responses | SithBot → Patient: return category recommendation + disclaimer.

4. Functional Requirements

Functional requirements are prioritised using the MoSCoW framework: M = Must have (MVP critical), S = Should have (high value), C = Could have (desirable), W = Won't have in first release. Each requirement references the source use case and includes a verification test case ID.

4.1 Authentication and Access Control

ID	Description	Priority	Source	Verification
FR-01	The system shall allow unregistered users to create a patient account using email or phone number with OTP verification.	M – Must	UC-01	TC-FR-01: Register with valid email; verify OTP; confirm account activated.
FR-02	The system shall enforce password complexity: minimum 8 characters, 1 uppercase, 1 digit, 1 special character.	M – Must	UC-01; NFR-S	TC-FR-02: Attempt registration with weak passwords; confirm rejection and error message.

FR-03	The system shall support role-based access control (RBAC) with roles: Patient, Doctor, Clinic Admin, Platform Admin, Volunteer, Moderator.	M – Must	UC-01; UC-15	TC-FR-03: Login as each role; verify different dashboards and menus are displayed.
FR-04	The system shall provide a secure password reset flow via email OTP expiring in 10 minutes.	M – Must	UC-01	TC-FR-04: Request reset; verify OTP expiry at 10 min; confirm password changed successfully.
FR-05	The system shall automatically log out users after 30 minutes of inactivity.	S – Should	NFR-S	TC-FR-05: Leave session idle for 31 minutes; confirm redirect to login page.
FR-06	The system shall support optional MultiFactor Authentication (MFA) for Clinic and Platform Administrators.	S – Should	NFR-S	TC-FR-06: Enable MFA; login from new device; confirm OTP prompt is displayed.

4.2 Profile and Consent Management

ID	Description	Priority	Source	Verification
FR-07	The system shall allow patients to update personal profile information including name, contact details, language preference, and profile photo.	M – Must	UC-02	TC-FR-07: Update profile fields; save; confirm changes persist on re-login.
FR-08	The system shall display privacy consent options and record consent acceptance and withdrawal with a timestamp.	M – Must	UC-02; PDPA	TC-FR-08: Withdraw consent; confirm timestamp recorded and data minimisation triggered.
FR-09	The system shall allow patients to add or update emergency contact name and phone number on their profile.	M – Must	UC-02; UC-10	TC-FR-09: Add emergency contact; trigger emergency flow; confirm contact details are shown.

FR-10	The system shall provide an Adult Mode with minimum 18px font, simplified navigation, and reduced feature set.	S – Should	UC-02; Persona 2	TC-FR-10: Enable Adult Mode; confirm menu reduced and text enlarged to 18px minimum.

4.3 Specialist Search and Discovery

ID	Description	Priority	Source	Verification
FR-11	The system shall allow patients to search for mental health specialists using filters: category, location, consultation type, language, and availability date.	M – Must	UC-03; Survey Q3	TC-FR-11: Apply all filters; confirm returned results match all applied criteria exactly.
FR-12	The system shall display specialist profiles with: name, photo, qualifications, SLMC/SLPA registration number, languages, consultation fee in LKR, rating, and available slots.	M – Must	UC-03	TC-FR-12: View specialist profile; confirm all specified fields are present and accurate.
FR-13	The system shall display clinic locations on an interactive map (Google Maps / OpenStreetMap).	S – Should	UC-03; Survey Q6	TC-FR-13: View clinic; confirm correct map pin and directions link.
ID	Description	Priority	Source	Verification
FR-14	The system shall allow patients to read and submit star ratings and text reviews for specialists, subject to moderation.	C – Could	UC-03; Competitive Analysis	TC-FR-14: Submit review; confirm displayed on profile after moderation approval.

4.4 Appointment Management

ID	Description	Priority	Source	Verification

FR-15	The system shall allow patients to book in-person or video appointments with available specialists.	M – Must	UC-04; Interview #1	TC-FR-15: Book appointment; confirm slot blocked and confirmation sent.
FR-16	The system shall allow patients to cancel appointments and trigger a refund if cancellation is more than 24 hours before the scheduled time.	M – Must	UC-04	TC-FR-16: Cancel 25+ hrs in advance; confirm refund initiated via PayHere.
FR-17	The system shall allow clinic administrators to approve, reject, or reschedule pending appointment requests.	M – Must	UC-06; Interview #2	TC-FR-17: Admin rejects booking; confirm patient receives notification with reason.
FR-18	The system shall provide an Emergency Same-Day Booking option when regular slots are full, subject to clinic approval.	S – Should	UC-04; UC-06	TC-FR-18: Attempt booking when full; confirm emergency option appears and admin notified.
FR-19	The system shall send SMS and email reminders to patients 24 hours and 1 hour before their appointment.	M – Must	UC-04; Survey Q7	TC-FR-19: Book appointment; verify two reminders are sent at the correct scheduled times.
FR-20	The system shall allow doctors to manage their weekly availability schedule (add, edit, and block slots).	M – Must	UC-05; Interview #2	TC-FR-20: Doctor blocks a slot; confirm slot is unavailable in-patient search immediately.

4.5 Telemedicine

ID	Description	Priority	Source	Verification

FR-21	The system shall generate a secure, time-limited video consultation room link for confirmed online appointments, available 15 minutes before the scheduled time.	M – Must	UC-07	TC-FR-21: Confirm online appointment; verify video link appears in dashboard 15 min before slot.
FR-22	The system shall allow the doctor to initiate the video session from the doctor dashboard.	M – Must	UC-07	TC-FR-22: Doctor clicks 'Start Session'; confirm video room opens and status updates.
FR-23	The system shall provide a one-tap phone call fallback option if the video connection fails.	S – Should	UC-07; Risk	TC-FR-23: Simulate video failure; confirm phone call fallback button appears.

4.6 Queue Management and Clinic Operations

ID	Description	Priority	Source	Verification
FR-24	The system shall assign a queue token to each confirmed in-person appointment and display the live queue position to patients.	M – Must	UC-09	TC-FR-24: Admin advances queue; confirm patient dashboard updates within 60 seconds.
FR-25	The system shall allow clinic administrators to advance, pause, and reset the daily patient queue.	M – Must	UC-09; Interview #2	TC-FR-25: Admin pauses queue; confirm patients see pause banner with expected resume time.
FR-26	The system shall send a push/SMS proximity notification when patient is within 3 tokens of their turn.	S – Should	UC-09	TC-FR-26: Advance queue to 3 tokens before patient; confirm notification is delivered.
FR-27	The system shall provide clinic administrators with a daily appointment list sortable by time, doctor, and status.	M – Must	UC-06; Interview #2	TC-FR-27: Login as clinic admin; verify list sorts correctly by all three criteria.

4.7 Payment Processing

ID	Description	Priority	Source	Verification
FR-28	The system shall integrate PayHere payment gateway for online appointment payments in LKR.	M – Must	UC-08	TC-FR-28: Complete online payment; confirm PayHere redirects and payment confirmed via webhook.
FR-29	The system shall allow clinic administrators to record and track cash payment status.	M – Must	UC-08; Interview #2	TC-FR-29: Mark appointment as cash paid; confirm payment record status updated to Paid.
FR-30	The system shall generate a PDF payment receipt for all confirmed transactions (online and cash).	M – Must	UC-08	TC-FR-30: Generate receipt; confirm PDF opens with correct appointment details, fee, and reference.
FR-31	The system shall process refunds for cancelled appointments via the original PayHere payment method.	S – Should	UC-08; FR-16	TC-FR-31: Cancel eligible appointment; confirm refund initiated and patient notified of status.

4.8 Emergency Support

ID	Description	Priority	Source	Verification
FR-32	The system shall provide a persistent emergency support button visible on all screens when a patient is logged in.	M – Must	UC-10	TC-FR-32: Login as patient; navigate multiple screens; confirm button is visible on all.
FR-33	The system shall require explicit user confirmation before initiating a crisis call or sharing location data.	M – Must	UC-10; PDPA	TC-FR-33: Press emergency button; confirm confirmation dialog appears with location consent prompt.

ID	Description	Priority	Source	Verification
FR-34	The system shall display Sri Lankan emergency helplines if the assigned doctor is unreachable within 20 seconds.	M – Must	UC-10	TC-FR-34: Simulate doctor unavailability; confirm helplines (1926, 0112 696 666, 0115 679 900) appear at 20s mark.
FR-35	The system shall display a nonsubstitution disclaimer: 'SithCare is not a substitute for emergency services. Dial 119 (Police) or 1990 (Ambulance).'	M – Must	UC-10; Risk	TC-FR-35: Open emergency screen; confirm disclaimer text is displayed prominently.
FR-36	The system shall log all crisis events with timestamp, anonymised patient ID, action taken, and outcome.	M – Must	UC-10; Risk	TC-FR-36: Trigger emergency; confirm log entry created and visible in Platform Admin panel.

4.9 AI Chatbot (SithBot)

ID	Description	Priority	Source	Verification
FR-37	The system shall provide SithBot, an AI triage chatbot accessible without login for basic mental health triage.	M – Must	UC-12	TC-FR-37: Access chatbot without login; confirm full triage flow completes and recommendation displayed.
FR-38	The system shall detect crisis keywords in chatbot input and immediately surface emergency support options, bypassing the triage flow.	M – Must	UC-12,Risk	TC-FR-38: Enter crisis keyword; confirm emergency options displayed within one chatbot response.
FR-39	The system shall display a nondiagnosis disclaimer on every SithBot response screen.	M – Must	UC-12, Risk	TC-FR-39: Complete triage; confirm disclaimer present on every screen, not only the final.

4.10 Community Features

ID	Description	Priority	Source	Verification
FR-40	The system shall provide an anonymous community forum where users post using pseudonymous platform identities.	S – Should	UC-11	TC-FR-40: Post anonymously; confirm real name is hidden from all other users.
FR-41	The system shall allow community moderators to flag, hide, and delete posts that violate community guidelines.	S – Should	UC-11	TC-FR-41: Moderator hides post; confirm removed from public view within 60 seconds.

4.11 Localisation and Communication

ID	Description	Priority	Source	Verification
FR-49	The system shall support English, Sinhala, and Tamil language switching accessible from any screen.	M – Must	Survey Q4; 1.2 Scope	TC-FR-49: Switch to Sinhala; confirm all key UI strings render correctly in Unicode Sinhala.
FR-50	The system shall display all monetary amounts in Sri Lankan Rupees (LKR).	M – Must	1.2 Scope	TC-FR-50: Complete booking flow; confirm all fees displayed as LKR values throughout.
FR-51	The system shall display local emergency helpline numbers (Sumithrayo 1926, CCCline 0112 696 666, Sahanaya 0115 679 900) in the emergency support section.	M – Must	UC-10; 1.2 Scope	TC-FR-51: Access emergency section; confirm all three numbers are displayed and each is tap-to-call.
FR-52	The system shall send appointment reminders and notifications via both SMS and email in the patient's preferred language.	M – Must	UC-04; Interview #1	TC-FR-52: Set language to Sinhala; book appointment; confirm both SMS and email arrive in Sinhala.

5. Non-Functional Requirements

5.1 Performance

ID	Requirement	Target / Threshold	Verification
NFR-P-01	Page load time on 4G LTE (>10 Mbps).	≤ 3 seconds for 95% of requests.	Performance test with JMeter / Lighthouse.
NFR-P-02	Appointment booking transaction (excluding PayHere redirect).	End-to-end ≤ 5 seconds.	Integration performance test.
NFR-P-03	Concurrent user support without degradation.	Minimum 500 concurrent users.	Load test with JMeter.
NFR-P-04	Live queue update propagation after admin action.	All connected patients updated within 60 seconds.	Automated queue advancement test.
NFR-P-05	AI chatbot (SithBot) response time.	≤ 2 seconds for 90% of queries.	Chatbot response timing test.

5.2 Security

ID	Requirement	Verification
NFR-S-01	All data in transit shall be encrypted using TLS 1.3.	SSL/TLS configuration review; automated SSL scan.
NFR-S-02	All patient health data at rest shall be encrypted using AES-256.	Database encryption configuration review.
NFR-S-03	JWT access tokens shall expire after 15 minutes; refresh tokens after 7 days.	Token expiry test: confirm expired token rejected.

ID	Requirement	Verification
NFR-S-04	The system shall implement OWASP Top 10 mitigations including input sanitisation, CSRF protection, and rate limiting.	OWASP security checklist review; penetration test.
NFR-S-05	Patient health/session records shall be accessible only to the authoring doctor and Platform Administrator.	RBAC access control test for each role.
NFR-S-06	Community forum posts shall use pseudonymous IDs; real user identity shall not be publicly linked.	Data exposure test; inspect forum post metadata.
ID	Requirement	Verification
NFR-S-07	Payment card data shall never be stored on SithCare servers; all payment data handled by PayHere (PCIDSS).	Architecture review; confirm no card data in SithCare DB.
NFR-S-08	All authentication events (login, logout, failed attempts) shall be logged with timestamp and IP address.	Log audit review; confirm log entries for all auth events.

5.3 Usability

ID	Requirement	Target / Threshold	Verification
NFR-U-01	System Usability Scale (SUS) score in usability testing.	≥ 70 (acceptable) in user testing sessions.	SUS questionnaire with test participants.
NFR-U-02	Core task completion time for new users (register, find doctor, book appointment).	Completable in under 5 minutes.	Timed usability test with 5 participants.
NFR-U-03	Adult Mode font size and navigation.	Minimum 18px font; single-level navigation menus.	Visual inspection and accessibility audit.
NFR-U-04	Screen reader / accessibility compliance.	WCAG 2.1 Level AA.	Accessibility audit using axe / WAVE tools.

NFR-U-05	Error messages across all userfacing screens.	Plain language in the user's selected language; no technical jargon.	Content review with native language reviewers.

5.4 Reliability

ID	Requirement	Target	Verification
NFR-R-01	System availability (excluding scheduled maintenance).	≥ 99.5% uptime.	Uptime monitoring (UptimeRobot or equivalent).
NFR-R-02	Scheduled maintenance windows.	Not during 08:00–22:00 Sri Lanka Standard Time (SLST).	Maintenance schedule review; advance notice to users.
NFR-R-03	Automated database backup frequency and recovery point.	Every 6 hours; RPO = 6 hours.	Backup log review; test restore from backup.
NFR-R-04	Emergency support subsystem availability during main application failure.	Emergency call/helpline feature must remain accessible even if non-critical modules fail.	Failover test: disable non-critical modules; confirm emergency feature operational.

5.5 Scalability

ID	Requirement	Verification
NFR-SC01	Application layer shall support horizontal scaling without code changes.	Architecture review; container/deployment configuration review.
NFR-SC02	Database schema-per-tenant design shall support up to 200 clinics without schema migration.	Tenant isolation test; load test with 200 simulated clinic schemas.

NFR-SC03	System shall handle a 5× traffic spike during public health campaigns.	Stress test: simulate 5× baseline load; confirm no downtime or degraded core function.

5.6 Maintainability

ID	Requirement	Verification
NFR-M-01	All code shall follow consistent coding standards documented in CONTRIBUTING.md.	Code review; linting pass in CI pipeline.
NFR-M-02	Unit test coverage for all backend service modules.	Minimum 70% line coverage reported by test runner.
NFR-M-03	All API endpoints shall be documented using OpenAPI 3.0 specification.	OpenAPI spec review; confirm all endpoints documented.
NFR-M-04	No hardcoded secrets in source code; environmentbased configuration.	Static analysis scan; confirm secrets in .env files excluded from repository.
NFR-M-05	All code changes via pull requests with at least one peer review before merge.	GitHub PR history review; confirm branch protection rules active.

6. Alternative Solutions and Feasibility Study

6.1 Alternative Solutions Considered

✞ Alternative A – Manual Phone / Walk-in Booking (Current State)

- Patients contact clinics manually or visit physically. This is simple but causes delays, limited transparency, no central guidance, no digital receipts, no queue tracking, poor support for remote/elderly users, and significant stigma barriers from phone-based booking overheard by family members. This does not address the identified problem.

✞ Alternative B – Adapt an Existing Open-Source Platform (e.g., OpenMRS)

- OpenMRS is an open-source electronic medical records system used in developing countries. It could be extended to support mental health appointments. However, it is primarily designed for in-person clinical workflows in hospital settings, lacks patient-facing consumer UX, has no native community or volunteer features, and would require substantial customisation to meet Sri Lankan mental health context requirements. The learning curve and configuration overhead were assessed as too high for an 8-week design phase.

✞ Alternative C – Build on a Healthcare SaaS Platform (e.g., Medplum)

- Medplum is a FHIR-compliant healthcare developer platform providing authentication, patient records, and scheduling APIs. It is optimised for the US healthcare market (HL7 FHIR, US insurance workflows). Sri Lankan context requirements (PayHere, Sinhala/Tamil localisation, local emergency helplines, LKR pricing) would require significant custom development on top of a complex proprietary stack. Vendor dependency and licensing costs are additional concerns for a university project.

✞ Alternative D – Social Media / Informal Support Groups (e.g., WhatsApp Groups)

- Users currently use public or private groups for community support, but privacy, misinformation, moderation, crisis handling, and professional connection are poor. This does not provide appointment management, verified specialist discovery, payment handling, or emergency support.

✞ Recommended Solution – Custom-Built Web Application (SithCare)

A custom-built solution using established open-source technologies (React, Node.js/FastAPI, PostgreSQL) provides full control over localisation, UX, data privacy, and compliance. It directly addresses the Sri Lankan context without the constraints of an existing platform and is achievable within the team's skill set and the SENG 31242 timeline.

6.2 Feasibility Assessment

Dimension	Assessment	Justification
Technical	Feasible	Team skills include React, Node.js, SQL, and REST APIs from prior coursework. Telemedicine SDK (Jitsi) is free and open-source. PayHere provides a developer sandbox for testing.
Economic	Feasible (Academic)	Development cost is team effort only. Cloud hosting can use free-tier (AWS Educate / DigitalOcean student credits). PayHere has no monthly fee for sandbox testing. SMS/email usage within free-tier limits during development.

Operational	Feasible	Clinic administrators are technically capable of using web dashboards per interview findings. Patient-facing UI designed for low-literacy users (Adult Mode, multi-language, guided flows).
Schedule	Feasible with MoSCoW	8-week design phase is sufficient for core MVP requirements (Auth, Booking, Emergency, Queue). Advanced features (AI chatbot, volunteer portal, community) are Could-have and deferred to the SENG 34213 development phase.
Legal / Ethical	Feasible with Constraints	Requires strict privacy controls, PDPA-awareness checklist, RBAC, data minimisation, consent recording, and clear disclaimers that SithCare is not an emergency service or diagnostic tool.

6.3 MoSCoW Scope Prioritisation

Priority	Features
Must Have	Authentication and privacy, role-based dashboards (Patient, Clinic Admin, Platform Admin), mental health specialist search and verified profiles, appointment booking with PayHere and cash payment, clinic/doctor availability management, admin user management, emergency support with helpline display and crisis logging, basic SMS/email notifications, multilingual support (Sinhala, Tamil, English).
Should Have	Telemedicine (video consultation with fallback), PDF receipt generation, live patient queue tracker, doctor verification workflow, content moderation tools for moderators, appointment approval/reschedule workflow, Adult Mode, multi-branch clinic management, appointment cancellation and refund.
Priority	Features
Could Have	Anonymous community consultation forum, volunteer listener portal, AI triage chatbot (SithBot), doctor session notes, recovery stories module, specialist ratings and reviews, analytics dashboard for Platform Admin, mood and progress tracking.

Won't Have (v1.0)	AI clinical diagnosis, automatic emergency service dispatch without user confirmation, integration with government EHR systems, native Android/iOS mobile applications, insurance claim processing, complex fundraising automation.

Appendix A – Risk and Mitigation Register

Risk	Likelihood	Impact	Mitigation Strategies
Video consultation latency / reliability in Sri Lanka	Low	Medium	Keep telemedicine optional; use video SDK with Southeast Asia servers; provide one-tap phone fallback; in-person booking always available.
AI chatbot medical liability or inaccurate triage	Low	High	AI used for triage/navigation only; prohibit clinical diagnosis; mandatory disclaimer on all chatbot screens; logic reviewed with mental health professional before deployment.
Scope too large for 8-week design phase	Medium	High	Apply MoSCoW; focus MVP on Auth, Booking, Emergency, Queue; advanced features to backlog; track via GitHub Projects sprint board.
Patient data privacy or confidentiality breach	Medium	High	AES-256 at rest; TLS 1.3 in transit; pseudonymous IDs for community; strict RBAC; audit logging; PDPA compliance checklist during design.
Crisis feature misuse or system failure during emergency	Low	Very High	Explicit user confirmation before emergency actions; location sharing opt-in per session; SithCare non-substitution disclaimer always displayed; static fallback helplines require no backend connection.
Volunteer listener burnout or misconduct	Medium	Medium	Mandatory mental health first-aid training; code of conduct signed before activation; 60-minute session limits; 'Report Concern' button in every session; moderator review queue.
PayHere integration failure or downtime	Low	High	Cash payment as permanent alternative; payment retry with 5-minute slot reservation; webhook retry mechanism; patient notified of payment status.

Appendix B – Glossary of Sri Lankan Mental Health Terms

Term	Definition
SLMC	Sri Lanka Medical Council. The statutory body responsible for registering medical practitioners.
SLPA	Sri Lanka Psychological Association. The professional body for psychologists in Sri Lanka.
Sumithrayo	A volunteer-based Sri Lankan suicide prevention hotline. Helpline: 1926.
CCCline	Community Counselling Centre helpline operated by the Ministry of Health. Helpline: 0112 696 666.
Sahanaya	A counselling and rehabilitation centre in Sri Lanka. Helpline: 0115 679 900.
PDPA	Personal Data Protection Act – proposed Sri Lankan data protection legislation governing how personal data is collected, stored, and processed.
SithBot	The AI triage chatbot component of SithCare. Provides guided specialist category recommendations; does not provide clinical diagnoses.
Adult Mode	A simplified interface mode within SithCare designed for elderly or low-digitalliteracy users, featuring larger text, reduced navigation complexity, and essential features only.

Appendix C – Planned Repository Artefacts

Repository Path	Purpose

Repository Path	Purpose
documents/srs/	SRS draft and final documents, use case descriptions, requirements tables.
documents/diagrams/use-case/	Use case diagram source (.drawio) and exports (SVG/PDF).
documents/diagrams/activity/	Activity diagram source and exports for all planned activity diagrams.
research/Patient_Content/Transcripts folder	Patient interview transcripts
research/Specialist_Content/Transcripts folder	Mental health specialist interview transcripts
research/Survey_Analysis folder	Insights of the genereal questionnaire used in the Google Forms survey.
research/competitive-analysis.md	Feature comparison matrix: BetterHelp, Talkspace, Wysa, NalandaCare, SithCare.
research/sithcare_research folder	Literature review notes covering WHO Atlas, MoH Policy, and Digital Health Blueprint.
research/Survey_Questionnaire folder	PDF which contains the general questions in the Google Survey form

Appendix D – Draft Completion Checklist

- Complete patient questionnaire response analysis and update problem/user-needs section.
- Complete mental health specialist interview and update professional workflow requirements.
- Review all functional requirements with supervisor; confirm priorities using MoSCoW and sprint capacity.

- Update references using final literature review evidence matrix.
- Confirm system sequence diagrams (SSD-01 to SSD-05) are finalised and added to repository.
- Run accessibility review using axe / WAVE against usability NFRs.
- Run formatting and grammar review before final SRS submission.

Appendix E – Revision History

Version	Date	Changes	Author(s)
0.1	2026-04-30	Initial outline, scope definition, and section skeleton.	Group 10
1.0	2026-05-03	Full SRS draft: all chapters, 15 use cases with complete descriptions, 52 functional requirements, NFRs, feasibility study. Formatted for Sprint 2 submission.	Group 10